

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“I still see too many UIs being designed and developed by developers, and that has to stop...”

Jayraj Ugarkar heads the Internet of Things Centre of Excellence at Infosys Labs. His focus is on building products and solutions, and identifying new opportunities in IoT technology. He spoke to Janani Gopalakrishnan Vikram about how smartphones are becoming the dominant field device in the IoT era

JANANI GOPALAKRISHNAN VIKRAM

What do you think are the main prerequisites for a device to be called ‘smart’ or ‘IoT-ready?’

Devices in the IoT are broadly classified as passive, active and autonomous. For a device to qualify as ‘IoT ready,’ it needs to be capable of at least communicating one of three parameters—status, performance and/or location. Both passive and active devices are capable of doing so after varying degrees of instrumentation. For example, a crate of fruits is passive, but can be made to communicate by instrumenting it with an RFID tag. However, to qualify as ‘smart,’ the device needs to be capable of making decisions locally, based on preset rules or conditions.

Can smartphones be considered the *de facto* gateway for IoT-devices to connect to the Internet—what change is this going to bring about in the phone?

Gateways play a very important role in the IoT world. They are the point where data aggregation, data processing and communication with the rest of the world happen. There is still some way to go before smartphones become the *de facto* gateway in the IoT world, and there is no doubt in my mind

that they will get there. The primary changes I expect are support for multiple protocols commonly used in the IoT world like Zigbee, NFC, Flutter, etc.

Will this trend bring about any significant changes in the UI of smartphones?

To succeed, the IoT concept needs a powerful, handy, easy to use and ubiquitous interface for humans to interact with things around them. I agree purists define the IoT as a world where machines interact amongst themselves, taking decisions and acting without human intervention. But the ultimate goal of enabling machines to take decisions and act is for humans to benefit. It is not for any other reason. There is no device better than a smartphone for humans to use whenever their intervention is required, or when they want to reap the benefits of IoT. However, smartphone ecosystems need to go through significant changes to meet these requirements. Besides supporting IoT protocols, lots of apps are needed to help us interact with our car, refrigerator, home security system, etc.

How long will the smartphone remain a popular interface for IoT products? When will devices become smart enough to do away with this interface too?

Rome was not built in a day! I

strongly believe devices will become more and more autonomous as we move along. However, the role of a smartphone as the interface between things and humans will remain significant. The reason is we have limited ability and patience to deal with multiple devices. It is like dealing with multiple remotes, which nobody wants. I don’t want to interact with my garage door, security alarm system, coffee machine, hot water system, etc, individually. When I come home or leave home, I want to have just one interface through which I can convey what I want each of the devices to do.

True, nobody wants to be surrounded by devices clamouring for their attention, but when devices do become autonomous, what should designers keep in mind when designing their UIs?

This is easier said than done, but simplicity in the UI is the key. I still see too many UIs being designed and developed, and that has to stop, especially when you are talking about a cutting-edge field and where the users involved will come from all walks of life. Designing human-machine interactions is part science and part art, and not to be treated as an afterthought.